

- N-channel enhancement mode MOSFET
- P-channel enhancement mode MOSFET
- N-channel depletion mode MOSFET
- P-channel depletion mode MOSFET

Depletion mode MOSFETs are normally on, and applying V_{GS} voltage turns them off. Enhancement mode MOSFETs are normally off, and applying V_{GS} voltage turns them on. The flow chart in Fig. 2 shows how different types of MOSFETs work.

Power MOSFET is the most used power semiconductor device in the world. It is an enhanced version of a regular MOSFET that is capable of handling significantly higher power levels compared to any other semiconductor device, such as an insulated-gate bipolar transistor (IGBT) and a thyristor.

A power MOSFET has significant advantages, which include high low-voltage efficiency and a high switching speed. It also has a good paralleling capability, that is, multiple MOSFETs of similar type can be connected parallelly to allow a much higher amount of current to flow through the setup.

Double-diffused MOSFET, commonly known as DMOS, is one of the most commonly used power MOSFETs. It is commonly used in high-voltage, high-frequency switching applications. The double diffusion process is used to form the source and a channel region of this MOSFET. Fig. 3 shows the cross-section of a power MOSFET.

Important MOSFET parameters

Two important parameters of a MOSFET that are crucial while selecting a MOSFET are the on-resistance, R_{DS(on)}, and the gate charge, Q_g. These two parameters are interrelated and are responsi-

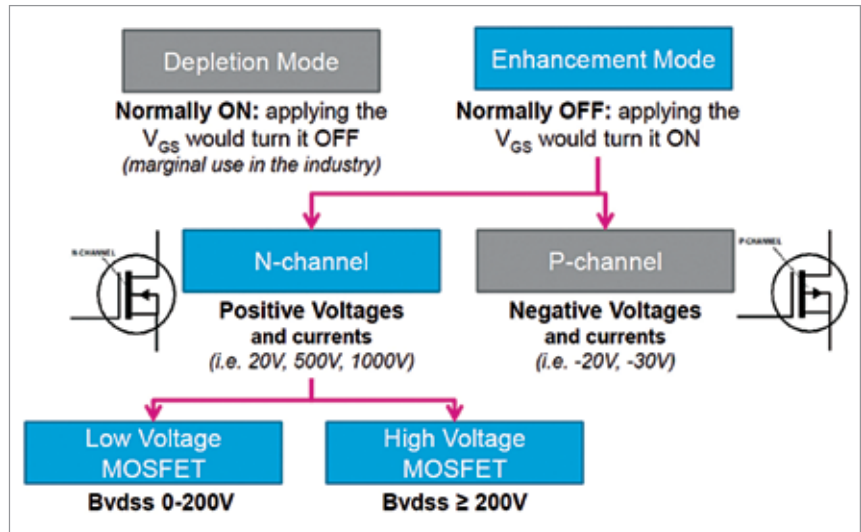


Fig. 2: Flow chart showing classification of MOSFETs (Source: STMicroelectronics)

ble for the efficiency of the system.

Some other important parameters that determine the performance of a MOSFET are the breakdown voltage, BV_{DSS}, and the body drain diode, which is essential to take into account when the device is used as a power diode, like in synchronous free-wheeling operation mode, and intrinsic capacitances that can affect the switching times and voltage spikes.

On-state resistance, R_{DS(on)}. It denotes the resistance between the drain and source terminals when the MOSFET is in on state. The conduction loss depends on it; the lower the value of the R_{DS(on)}, lower is the conduction loss.

Total gate charge, Q_g. It denotes the electric charge required by the gate driver to turn the device on/off.

Figure of Merit, FoM. It is the product of R_{DS(on)} and Q_g, which accounts for the conduction losses and switching losses of a MOSFET. Therefore, the efficiency of a MOSFET depends on both, the R_{DS(on)} and the Q_g.

Breakdown voltage, BV_{DSS}. It denotes the maximum drain-to-source

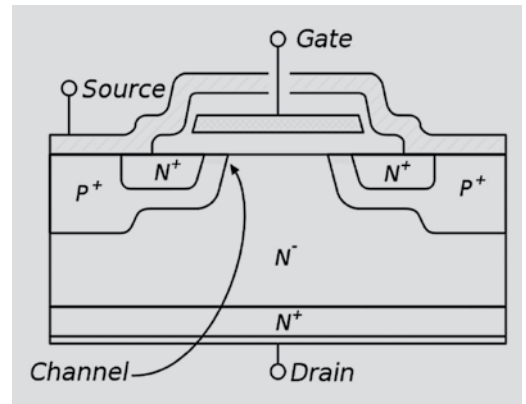


Fig. 3: Cross-section of a power MOSFET (Source: Link)

voltage a MOSFET is capable of sustaining in off state.

An example of a MOSFET's datasheet parameters is given in Fig. 4.

Selecting a power MOSFET

Selecting the right power MOSFET with correct characteristics and appropriate ratings is important for designing a great circuit. The device should have operating conditions or working range that tells us the safe conditions for its working. The absolute maximum ratings are the limiting values, exceeding which might cause permanent damage or failure of the circuit.

"Selection of a MOSFET is very crucial for the stability of a sys-